

4.1 Analytical Techniques

Question Paper

Course	CIEA Level Chemistry
Section	4. Analysis
Topic	4.1 Analytical Techniques
Difficulty	Easy

Time allowed: 30

Score: /23

Percentage: /100

Question 1a

There are three possible isomers, **A**, **B** and **C**, with the molecular formula $\text{C}_3\text{H}_8\text{O}$.

Isomers **A** and **B** are positional isomers of each other.

Isomer **C** is a functional group isomer of **A** and **B**.

A student uses the following equation to calculate that the percentage by mass of carbon in all three isomers is 60.0%.

$$\text{Percentage by mass} = \frac{\text{total mass of element}}{\text{overall mass of compound}} \times 100$$

Calculate the percentage by mass of hydrogen and oxygen in all three isomers.

[2 marks]

Question 1b

The infrared spectrum of isomer **C** shown in Fig. 1.1 does not have a peak in the region of $3200 - 3600 \text{ cm}^{-1}$.

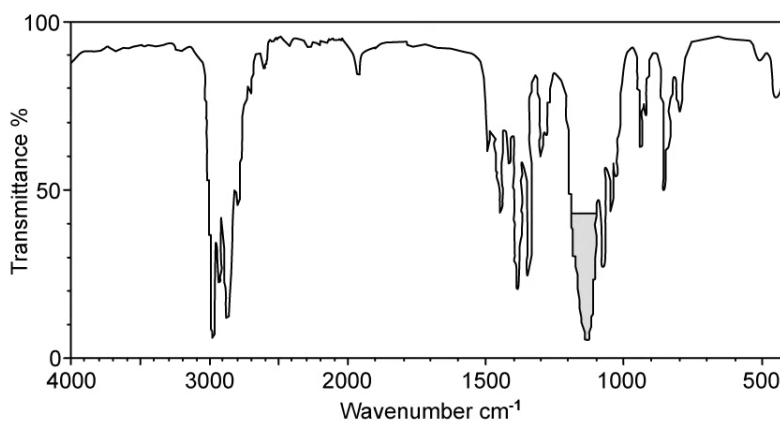


Fig. 1.1

i)

Using the infrared spectrum in Fig. 1.1 and Table 1.1, suggest a structure for isomer **C**.

Table 1.1

Bond	Functional groups containing the bond	Characteristic infrared absorption range (in wavenumber) / cm^{-1}
C–O	hydroxy, ester	1040 – 1300
C=C	aromatic compound, alkene	1500 – 1680
C=O	amide	1640 – 1690
	carbonyl, carboxyl	1670 – 1740
	ester	1710 – 1750
C≡N	nitrile	2200 – 2250
C–H	alkane	2850 – 2950
N–H	amine, amide	3300 – 3500
O–H	carboxyl	2500 – 3000
	hydroxy	3200 – 3600

[1]

ii)

Suggest which bonds the highlighted peak at around 1130 cm^{-1} could represent.

[1]

[2 marks]

Question 1c

Isomers **A** and **B** both have a peak at $3200 - 3600\text{ cm}^{-1}$ in their infrared spectra.

State **two** possible chemical names for isomers **A** and **B**.

[1 mark]

Question 1d

Isomer **A** has a major peak on its mass spectrum at $m/e = 31$.

Suggest, with a reason, which compound is isomer **A**.

[2 marks]

Question 2a

Mass spectroscopy is an analytical technique that can be used to analyse elements and compounds.

Using mass spectroscopy, a sample of boron was found to contain two isotopes ^{10}B and ^{11}B with a relative abundance of 20% and 80% respectively.

Calculate the relative atomic mass of boron. Show your working.

[2 marks]

Question 2b

Fig. 2.1 represents the mass spectrum of potassium.

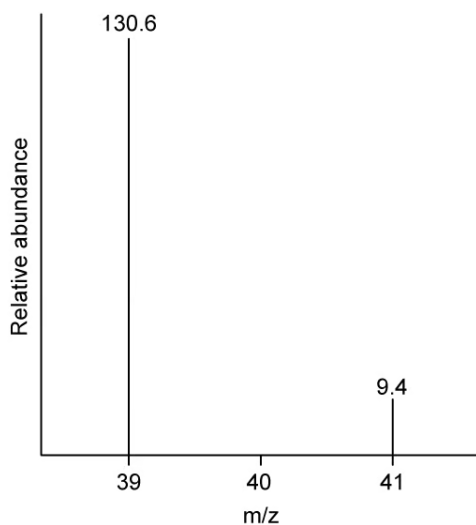


Fig. 2.1

Use the information in Fig 2.1 to calculate the relative atomic mass of potassium. Show your working.

[2 marks]

Question 2c

The mass spectra of octane and butane were obtained and analysed.

Write the equation for the formation of the molecular ion of octane and predict its m/e value.

[2 marks]

Question 2d

The mass spectrum of butane shows a molecular ion peak at $m/e = 58.0$.

Explain why there is also a smaller peak at $m/e = 59.0$ on the mass spectrum of butane.

[2 marks]

Question 2e

State, with a reason, whether butane or octane will have a higher $[M+1]$ peak.

[1 mark]

Question 2f

State what the difference would be in the ratio of the peak heights of the M^+ peak to the $[M+2]$ peak in 1-bromobutane and 1-chlorobutane.

[2 marks]

Question 3a

This question is about some isomeric alcohols with the molecular formula $C_5H_{12}O$.

Some alcohols were heated with potassium dichromate(VI) and sulfuric acid. The organic compounds were separated from the reaction mixtures and purified.

The infrared spectra of two of these organic compounds are shown in Fig. 3.1.

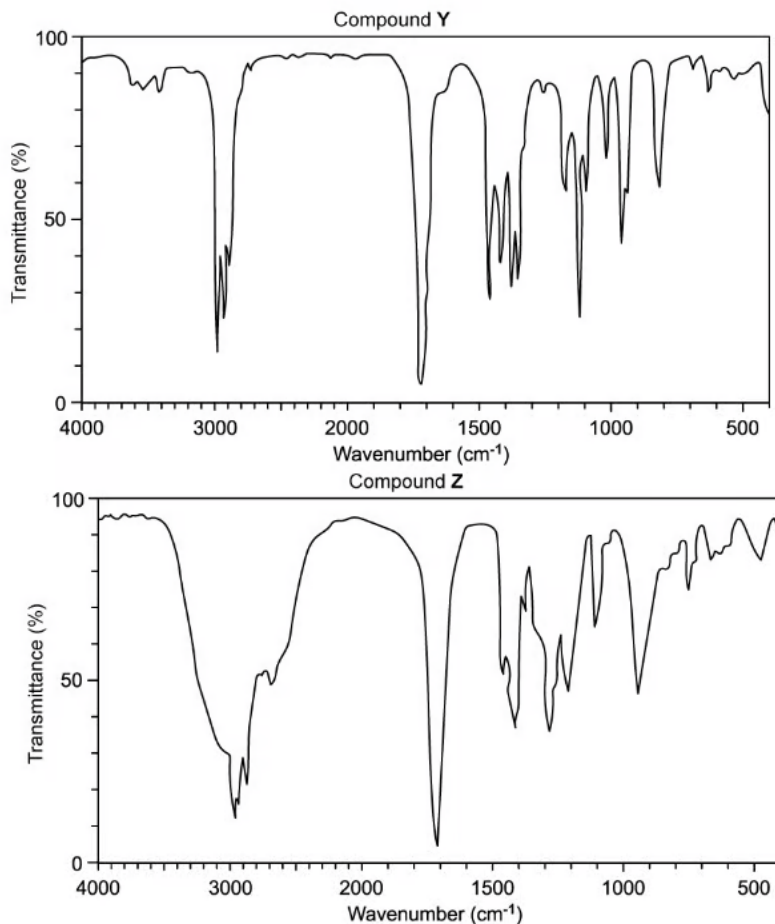


Fig. 3.1

Using the data in Table 3.1, deduce the type of compound responsible for each spectrum.

Table 3.1

Bond	Functional groups containing the bond	Characteristic infrared absorption range (in wavenumber) / cm^{-1}
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C-O	hydroxy, ester	1040 – 1300
C=C	aromatic compound, alkene	1500 – 1680
C=O	amide	1640 – 1690
	carbonyl, carboxyl	1670 – 1740
	ester	1710 – 1750
C≡N	nitrile	2200 – 2250
C-H	alkane	2850 – 2950
N-H	amine, amide	3300 – 3500
O-H	carboxyl	2500 – 3000
	hydroxy	3200 – 3600

Your answer should include references to wavenumbers and their corresponding bonds.

[2 marks]

Question 3b

The infrared spectrum a student obtained of an organic molecule which contains carbon, hydrogen and oxygen atoms is shown in Fig. 3.2.

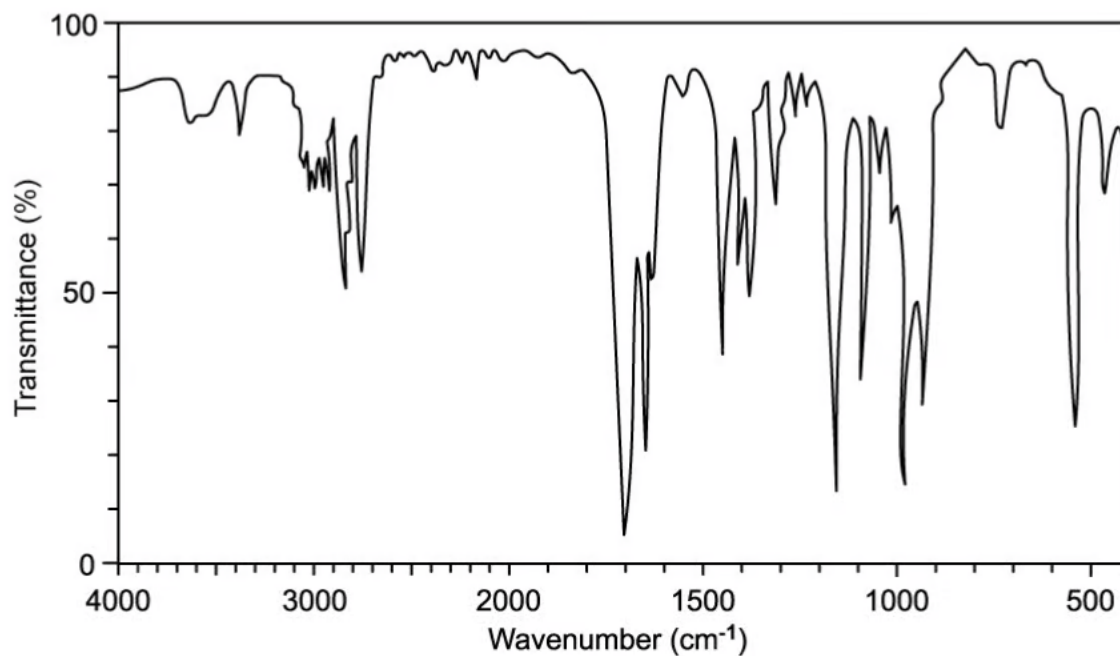


Fig. 3.2

Use Table 3.1 in part (a) and information from the infrared spectrum to explain how the student deduced that the spectrum shows the presence of a carbonyl group.

[1 mark]

Question 3c

In an experiment to prepare a sample of ethanal, CH_3CHO , ethanol, $\text{C}_2\text{H}_5\text{OH}$, is reacted with acidified potassium dichromate(VI) and the reaction mixture is distilled. The infrared spectra for ethanol and ethanal are shown in Fig. 3.3.

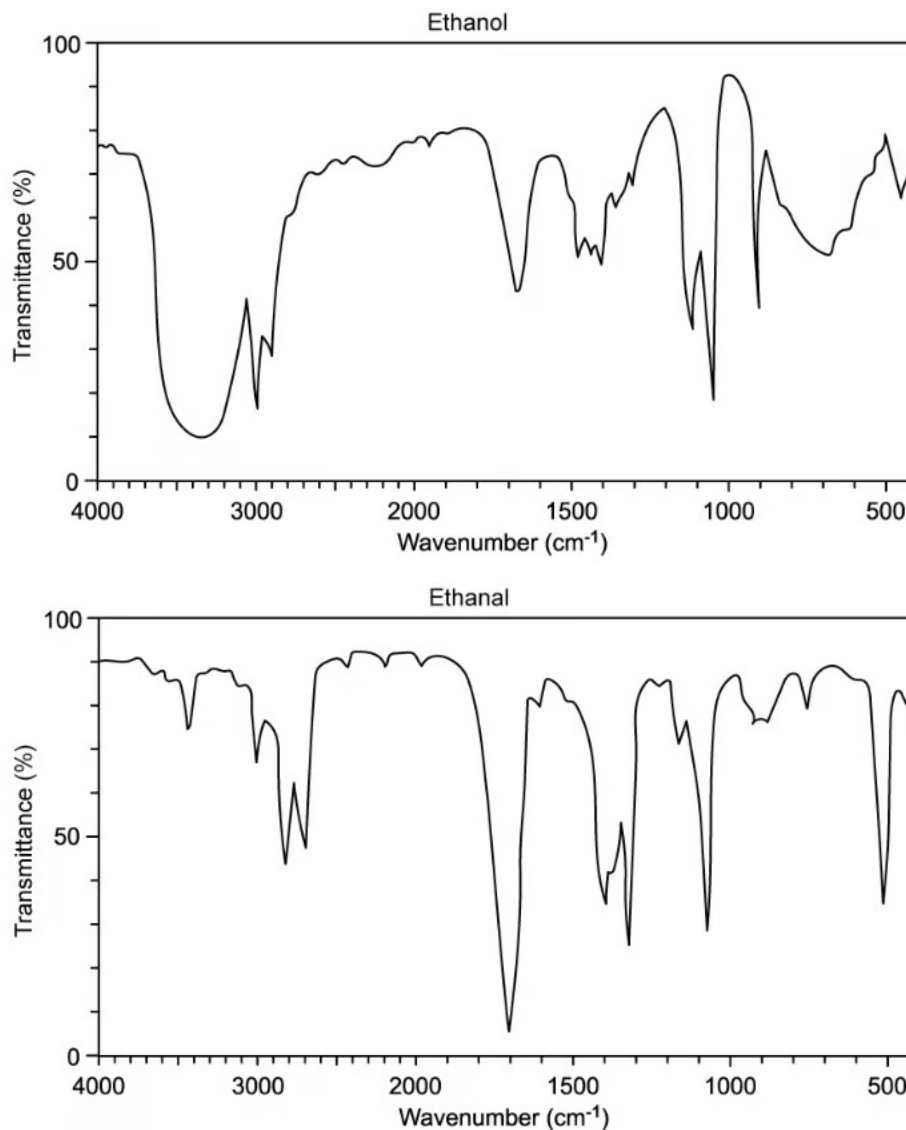


Fig. 3.3

i)

State the bonds that give rise to the absorption in the ethanol spectrum at 3400 cm^{-1} and the absorption in the ethanal spectrum at 1720 cm^{-1} .

[1]

ii)

Explain why the absorption at 3400 cm^{-1} in the ethanol spectrum does not appear in the spectrum for ethanal.

[1]

[2 marks]

Model Answers: Easy

1a

a) The percentage by mass of hydrogen and oxygen in all three isomers is:

- M_r of $C_3H_8O = (3 \times 12.0) + (8 \times 1.0) + 16.0 = \underline{60.0}$; [1 mark]
- Percentage by mass of hydrogen = $\frac{8.0}{60.0} \times 100 = 13.3\%$

AND

$$\text{Percentage by mass of oxygen} = \frac{16.0}{60.0} \times 100 = 26.7\%; \text{ [1 mark]}$$

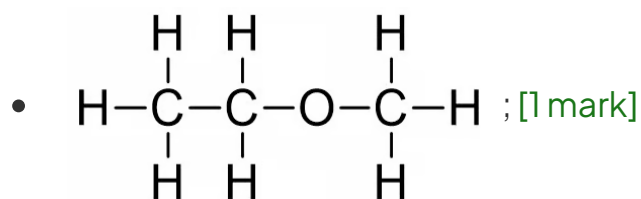
[Total: 2 marks]

- Isomers will always have the same percentage by mass data for each element
- When examiners are writing questions about any compound, they can use percentage by mass for each element within the compound to create the information that you need for an empirical formula calculation

1b

b)

i) Using the infrared spectrum, a possible structure for isomer **C** is:



ii) The highlighted peak at around 1130 cm^{-1} could represent the:

- C-O bonds; [1 mark]

[Total: 2 marks]

- The chemical formula C_3H_8O suggests that the structure is saturated (all single bonds)
- You are told that compound **C** does not have a peak at $3200 - 3600\text{ cm}^{-1}$ which corresponds to the hydroxy group
- With no O-H peak, there is only one possible structure that has the correct number of hydrogens
 - This is an example of an ether
 - Whilst ethers are not expected knowledge, they can be included as possible options for isomers
- Since there is no O-H peak, the only realistic option for the peak at around 1130 cm^{-1} is the C-O bonds
 - Again, you are not expected to work in the fingerprint region however you can be asked to infer answers from the fingerprint region with sufficient information

1c

c) **Two** possible chemical names for isomers **A** and **B** are:

- Propan-1-ol
AND
Propan-2-ol; [1 mark]

[Total: 1 mark]

- In the opening section of this entire question, you are told that **A** and **B** are positional isomers of each other
- The peak at $3200 - 3600\text{ cm}^{-1}$ indicates that the hydroxyl group is present in both compounds
- This should inform you that **A** and **B** are both alcohols
- This means that there are only two possibilities for **A** and **B** which are propan-1-ol and propan-2-ol
 - You do not have the information to distinguish between **A** and **B**

1d

d) Isomer **A** is:

- Propan-1-ol; [1 mark]

Explanation:

- The peak at $m/e = 31$ is due to CH_2OH^+ ; [1 mark]

[Total: 2 marks]

- You would be awarded the final mark if you gave CH_2OH as your answer but the peaks are due to positive ions so it is good practice to add the + sign
- If you think about breaking the C-C bonds within each structure
 - Propan-1-ol gives:
 - CH_3 group with a mass of 15
 - CH_2 group with a mass of 14
 - CH_2OH group with a mass of 31
 - You can also get combinations of these groups
 - CH_3CH_2 group with a mass of 29
 - $\text{CH}_2\text{CH}_2\text{OH}$ group with a mass of 45
 - Propan-2-ol gives:
 - CH_3 group with a mass of 15 - the peak could potentially be twice as large as there are 2 methyl groups
 - CHOH group with a mass of 30
 - You can also get one combination from these groups
 - CH_3CHOH group with a mass of 45

2a

a) To calculate the relative atomic mass of boron:

- $A_r\text{B} = \frac{(10 \times 20) + (11 \times 80)}{100}$; [1 mark]
- $A_r\text{B} = 10.8$; [1 mark]

[Total: 2 marks]

- The relative atomic mass of an element can be calculated by using the relative abundance values using the following equation:

$$\frac{(\text{relative abundance}_{\text{isotope 1}} \times \text{mass}_{\text{isotope 1}}) + (\text{relative abundance}_{\text{isotope 2}} \times \text{mass}_{\text{isotope 2}}) \text{ etc}}{100}$$

- The relative abundance of an isotope is either given or can be read off the mass spectrum

2b

b) To calculate the relative atomic mass of potassium:

- Total abundance = $130.6 + 9.4 = 140$; [1 mark]
- $A_r \text{ K} = \frac{(39 \times 130.6) + (41 \times 9.4)}{140} = 39.1 / 39.14$; [1 mark]

[Total: 2 marks]

- **Careful:** Abundance can be given as a percentage or the abundance of the atoms
- Since the data can be presented to you in different ways, take your time to make sure you have the correct values in the correct place in the equation
 - In this calculation, you are using information from the mass spectrum which shows the relative abundance
 - Therefore, you need to use 140 as the total abundance, not 100%

2c

c) The equation for the formation of the molecular ion of octane and its m/e value are:

- $\text{C}_8\text{H}_{18} \rightarrow \text{C}_8\text{H}_{18}^+ + \text{e}^-$; [1 mark]
- $m/e = (8 \times 12) + (18 \times 1) = 114$; [1 mark]

[Total: 2 marks]

- The molecular ion is formed by losing one electron
- Since the mass of an electron is negligible, the molecular ion will have the same mass as the molecule itself

2d

d) There is also a smaller peak at $m/e = 59.0$ on the mass spectrum of butane because of:

- Carbon-13 / ^{13}C isotopes; [1 mark]
- Which adds a mass of 1 onto the molecular ion / causes an $M + 1$ peak; [1 mark]

[Total: 2 marks]

- Most carbon atoms are carbon-12

- However, approximately 1.1% of all carbon atoms are carbon-13
- This means that there is a chance that one (or potentially more) of the carbon atoms in butane will have a mass of 13 instead of 12

2e

e) The alkane that will have a higher $[M+1]$ peak is:

- Octane

AND

(Because,) it contains more carbon atoms; [1 mark]

[Total: 1 mark]

- The $M+1$ peak appears due to the presence of carbon-13 atoms
- Since octane has more carbon atoms, there is a greater chance of one atom being carbon-13 which results in a larger $M+1$ peak

2f

e) The difference in the ratio of the M^+ peak to the $[M+2]$ peak in 1-bromobutane and 1-chlorobutane would be:

- In 1-chlorobutane, the ratio of M^+ : $[M+2]$ would be 3:1; [1 mark]
- In 1-bromobutane, the ratio of M^+ : $[M+2]$ would be 1:1; [1 mark]

[Total: 2 marks]

- Both chlorine and bromine exist as two isotopes:
 - ^{35}Cl and ^{37}Cl
 - ^{35}Cl is 3 times more abundant than ^{37}Cl
 - ^{79}Br and ^{81}Br
 - Their relative abundances are the same
- Compounds containing one Cl atom will have an M^+ peak due to the ^{35}Cl isotope and a $[M+2]$ peak due to ^{37}Cl
 - The M^+ peak is 3 times higher than the $[M+2]$ peak
 - This is because the ^{35}Cl isotope is 3 times more abundant than ^{37}Cl
- Compounds containing one Br atom will have an M^+ peak due to the ^{79}Br isotope and a $[M+2]$ peak due to ^{81}Br

- The M^+ and the $[M+2]$ peak heights are the same
- This is because the abundances of the isotopes are the same

3a

a) The type of compound responsible for each spectrum is:

- **Y** = ketone / aldehyde

AND

The spectrum contains an absorption at $1700 - 1750 \text{ cm}^{-1}$ for the C=O bond / carbonyl group; [1 mark]

- **Z** = carboxylic acid

AND

The spectrum contains absorption at $2500 - 3000 \text{ cm}^{-1}$ for the O-H bond in acids and $1700 - 1750 \text{ cm}^{-1}$ for the C=O bond; [1 mark]

[Total: 2 marks]

- Make sure you quote **both** the absorption wavenumbers **AND** the bonds responsible

[Total: 2 marks]

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The spectrum contains an absorption at $1700 - 1750 \text{ cm}^{-1}$ for the $\text{C}=\text{O}$ bond / carbonyl group; [1 mark]

- **Z** = carboxylic acid

AND

The spectrum contains absorption at $2500 - 3000 \text{ cm}^{-1}$ for the $\text{O}-\text{H}$ bond in acids and $1700 - 1750 \text{ cm}^{-1}$ for the $\text{C}=\text{O}$ bond; [1 mark]

[Total: 2 marks]

- Make sure you quote **both** the absorption wavenumbers **AND** the bonds responsible
 - If you do not give both pieces of information then you are not giving enough detail to be awarded the mark
- Compound **Y** can be either a ketone or aldehyde
 - The question does not give enough information to be able to distinguish between them

3b

b) The infrared spectrum confirms the presence of a carbonyl group because:

- Absorption between $1700 - 1750 \text{ cm}^{-1}$ for the C=O bond; [1 mark]

[Total: 1 mark]

- **Remember:** A carbonyl group C=O bond is present in aldehydes, ketones and carboxylic acids

3c

c)

i) The bonds that give rise to the absorption in the ethanol spectrum at 3400 cm^{-1} and the absorption in the ethanal spectrum at 1720 cm^{-1} are:

- Ethanol, O-H (3400 cm^{-1})
AND
Ethanal, C=O (1720 cm^{-1}); [1 mark]

ii) The absorption at 3400 cm^{-1} in the ethanol spectrum does not appear in the spectrum for ethanal because:

- There is no O-H bond in ethanal / aldehydes; [1 mark]

[Total: 2 marks]

- It's important to remember the functional groups of your organic molecules and the key indicators you would expect to see on an infrared spectrum
- You need to be confident when using the wavenumbers in the data booklet to identify the different absorptions of these functional groups